MSG No. 67.502 $Cmma1'$ [Type II, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2221'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	`[1,2,3,4,17,18,19,20]`
2	$\left[\frac{3}{4}, 0, 0\right]$	`[5,6,7,8,21,22,23,24]`
3	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[9,10,11,12,25,26,27,28]`
4	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[13,14,15,16,29,30,31,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `2221'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[1,2,3,4,17,18,19,20]`
2	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[5,6,7,8,21,22,23,24]`
3	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,25,26,27,28]`
4	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,15,16,29,30,31,32]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m..1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,5,6,17,18,21,22]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[3,4,7,8,19,20,23,24]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,13,14,25,26,29,30]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[11,12,15,16,27,28,31,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `2/m..1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,5,6,17,18,21,22]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,4,7,8,19,20,23,24]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,13,14,25,26,29,30]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[11,12,15,16,27,28,31,32]`

Table 5: Wyckoff site: `4e`, site symmetry: `.2/m.1'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,3,13,15,17,19,29,31]`
2	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[2,4,14,16,18,20,30,32]`
3	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[5,7,9,11,21,23,25,27]`
4	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[6,8,10,12,22,24,26,28]`

Table 6: Wyckoff site: `4f`, site symmetry: `.2/m.1'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,3,13,15,17,19,29,31]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,4,14,16,18,20,30,32]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[5,7,9,11,21,23,25,27]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[6,8,10,12,22,24,26,28]`

Table 7: Wyckoff site: `4g`, site symmetry: `mm21'`

No.	position	mapping
1	$\left[0, \frac{1}{4}, z\right]$	`[1,6,12,15,17,22,28,31]`
2	$\left[0, \frac{3}{4}, -z\right]$	`[2,5,11,16,18,21,27,32]`
3	$\left[\frac{1}{2}, \frac{1}{4}, -z\right]$	`[3,8,10,13,19,24,26,29]`
4	$\left[\frac{1}{2}, \frac{3}{4}, z\right]$	`[4,7,9,14,20,23,25,30]`

Table 8: Wyckoff site: `8h`, site symmetry: `2..1'`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2,17,18]`
2	$\left[\frac{1}{2} - x, 0, 0\right]$	`[3,4,19,20]`
3	$[-x, 0, 0]$	`[5,6,21,22]`
4	$\left[x + \frac{1}{2}, 0, 0\right]$	`[7,8,23,24]`
5	$\left[x + \frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,25,26]`
6	$\left[-x, \frac{1}{2}, 0\right]$	`[11,12,27,28]`
7	$\left[\frac{1}{2} - x, \frac{1}{2}, 0\right]$	`[13,14,29,30]`
8	$\left[x, \frac{1}{2}, 0\right]$	`[15,16,31,32]`

Table 9: Wyckoff site: `8i`, site symmetry: `2..1'`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,2,17,18]`
2	$\left[\frac{1}{2} - x, 0, \frac{1}{2}\right]$	`[3,4,19,20]`
3	$\left[-x, 0, \frac{1}{2}\right]$	`[5,6,21,22]`
4	$\left[x + \frac{1}{2}, 0, \frac{1}{2}\right]$	`[7,8,23,24]`
5	$\left[x + \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,25,26]`
6	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	`[11,12,27,28]`
7	$\left[\frac{1}{2} - x, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,29,30]`
8	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	`[15,16,31,32]`

Table 10: Wyckoff site: `8j`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[\frac{1}{4}, y, 0\right]$	`[1,3,17,19]`
2	$\left[\frac{1}{4}, -y, 0\right]$	`[2,4,18,20]`

continued ...

Table 10

No.	position	mapping
3	$\left[\frac{3}{4}, -y, 0\right]$	[5,7,21,23]
4	$\left[\frac{3}{4}, y, 0\right]$	[6,8,22,24]
5	$\left[\frac{3}{4}, y+\frac{1}{2}, 0\right]$	[9,11,25,27]
6	$\left[\frac{3}{4}, \frac{1}{2}-y, 0\right]$	[10,12,26,28]
7	$\left[\frac{1}{4}, \frac{1}{2}-y, 0\right]$	[13,15,29,31]
8	$\left[\frac{1}{4}, y+\frac{1}{2}, 0\right]$	[14,16,30,32]

Table 11: Wyckoff site: 8k, site symmetry: .2.1'

No.	position	mapping
1	$\left[\frac{1}{4}, y, \frac{1}{2}\right]$	[1,3,17,19]
2	$\left[\frac{1}{4}, -y, \frac{1}{2}\right]$	[2,4,18,20]
3	$\left[\frac{3}{4}, -y, \frac{1}{2}\right]$	[5,7,21,23]
4	$\left[\frac{3}{4}, y, \frac{1}{2}\right]$	[6,8,22,24]
5	$\left[\frac{3}{4}, y+\frac{1}{2}, \frac{1}{2}\right]$	[9,11,25,27]
6	$\left[\frac{3}{4}, \frac{1}{2}-y, \frac{1}{2}\right]$	[10,12,26,28]
7	$\left[\frac{1}{4}, \frac{1}{2}-y, \frac{1}{2}\right]$	[13,15,29,31]
8	$\left[\frac{1}{4}, y+\frac{1}{2}, \frac{1}{2}\right]$	[14,16,30,32]

Table 12: Wyckoff site: 8l, site symmetry: ..21'

No.	position	mapping
1	$\left[\frac{1}{4}, 0, z\right]$	[1,4,17,20]
2	$\left[\frac{1}{4}, 0, -z\right]$	[2,3,18,19]
3	$\left[\frac{3}{4}, 0, -z\right]$	[5,8,21,24]
4	$\left[\frac{3}{4}, 0, z\right]$	[6,7,22,23]
5	$\left[\frac{3}{4}, \frac{1}{2}, z\right]$	[9,12,25,28]
6	$\left[\frac{3}{4}, \frac{1}{2}, -z\right]$	[10,11,26,27]
7	$\left[\frac{1}{4}, \frac{1}{2}, -z\right]$	[13,16,29,32]
8	$\left[\frac{1}{4}, \frac{1}{2}, z\right]$	[14,15,30,31]

Table 13: Wyckoff site: 8m, site symmetry: m..1'

No.	position	mapping
1	$[0, y, z]$	[1,6,17,22]
2	$[0, -y, -z]$	[2,5,18,21]
3	$\left[\frac{1}{2}, y, -z\right]$	[3,8,19,24]
4	$\left[\frac{1}{2}, -y, z\right]$	[4,7,20,23]
5	$\left[\frac{1}{2}, y+\frac{1}{2}, z\right]$	[9,14,25,30]
6	$\left[\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[10,13,26,29]
7	$\left[0, y+\frac{1}{2}, -z\right]$	[11,16,27,32]
8	$\left[0, \frac{1}{2}-y, z\right]$	[12,15,28,31]

Table 14: Wyckoff site: `8n`, site symmetry: `.m.1'`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ z\right]$	`[1,15,17,31]`
2	$\left[x,\ \frac{3}{4},\ -z\right]$	`[2,16,18,32]`
3	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ -z\right]$	`[3,13,19,29]`
4	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ z\right]$	`[4,14,20,30]`
5	$\left[-x,\ \frac{3}{4},\ -z\right]$	`[5,11,21,27]`
6	$\left[-x,\ \frac{1}{4},\ z\right]$	`[6,12,22,28]`
7	$\left[x+\frac{1}{2},\ \frac{3}{4},\ z\right]$	`[7,9,23,25]`
8	$\left[x+\frac{1}{2},\ \frac{1}{4},\ -z\right]$	`[8,10,24,26]`

Table 15: Wyckoff site: `16o`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,17]`
2	$[x,\ -y,\ -z]$	`[2,18]`
3	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	`[3,19]`
4	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	`[4,20]`
5	$[-x,\ -y,\ -z]$	`[5,21]`
6	$[-x,\ y,\ z]$	`[6,22]`
7	$\left[x+\frac{1}{2},\ -y,\ z\right]$	`[7,23]`
8	$\left[x+\frac{1}{2},\ y,\ -z\right]$	`[8,24]`
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[9,25]`
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[10,26]`
11	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	`[11,27]`
12	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	`[12,28]`
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	`[13,29]`
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ z\right]$	`[14,30]`
15	$\left[x,\ \frac{1}{2}-y,\ z\right]$	`[15,31]`
16	$\left[x,\ y+\frac{1}{2},\ -z\right]$	`[16,32]`